AI powered speech-to-speech mock interview platform

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1. Functional Requirements

The specific features the system provides to the user.

1.1 Resume Upload & Parsing

- FR 1: The system shall allow a user to upload a resume file in PDF or DOCX format.
- FR 2: The system shall extract raw text content from the uploaded resume.
- FR 3: The system shall parse the extracted text into structured fields, including skills, projects, work experience, and education.
- FR 4: The system shall display an error message if the uploaded file cannot be parsed (e.g. corrupted file, unsupported format).

1.2 Question Generation

- FR 5: The system shall generate a set of interview questions based on the structured resume data.
- FR 6: The system shall include resume-specific questions (referencing named projects/skills).
- FR 7: The system shall generate a fixed number of questions per session (configurable, default 6–8).

1.3 Speech Interaction

- *FR 8: The system shall convert each generated question into spoken audio and play it to the user.*
- *FR 9: The system shall allow the user to record a spoken response via the browser's microphone.*
- *FR 10: The system shall transcribe the recorded audio response into text.*
- *FR 11: The system shall advance to the next question only after the current response has been processed.*

1.4 Evaluation & Feedback

- *FR12: The system shall evaluate each transcribed response against a defined rubric (clarity, relevance, communication quality).*
- *FR13: The system shall generate a per-question score and short qualitative feedback.*
- *FR14: The system shall compile a final session report summarising all questions, answers, scores, and feedback.*
- *FR15: The system shall allow the user to view or download the final session report.*

1.5 Session Management

- *FR16: The system shall maintain session state for the duration of an interview session.*
- *FR17: The system shall allow a user to start a new session at any time.*
- *FR18: The system shall store completed session data (transcripts, scores) for later reference.*

2. Non-Functional Requirements

Non-functional requirements describe how the system should perform its functions — qualities such as speed, reliability, and usability rather than specific features.

Category	Requirement
Performance	*The end-to-end turnaround per question (speech-to-text, processing, text-to-speech) should complete within approximately 5–10 seconds on the target hardware to keep the interview flow usable.*
Usability	*The interface shall be simple enough for a first-time user to complete a full session without instructions beyond on-screen prompts.*
Reliability	*The system shall handle a failed transcription or generation call gracefully (retry or clear error message) without crashing the session.*
Availability	*The system shall be available and functional for the duration of a single interview session without requiring a stable long-term server connection (local-first design).*
Security	*Uploaded resumes and recorded audio shall not be shared with third parties beyond what is required for processing, and shall be deletable by the user.*
Maintainability	*The system shall be built as modular, independently testable components (parsing, question generation, speech, evaluation) to simplify future maintenance and upgrades*
Cost	*The system shall be operable at zero or near-zero recurring cost using free/open-source or freetier tools for the MVP version.*
Compatibility	*The frontend shall function correctly on current versions of major browsers (Chrome, Edge, Firefox, Safari) that support the MediaRecorder API.*

3. Literature Survey

The following works were reviewed to understand the current landscape of AI-driven interview preparation, resume analysis, and the core speech/language technologies this project depends on.

Title / Source	Summary	Relevance to This Project
https://skill-edge-zeta.vercel.app/	*Browser-native platform integrating resume scoring, job matching, skill gap identification, and personalised AI interview question generation using a Groq accelerated LLaMA inference layer.*	*Closest existing system to this project; validates the resume-to-question-generation approach and the use of fast, low-cost LLM inference.*
https://link.springer.com/chapter/10.1007/978-3-032-22911-3_6	*Combines NLP, multimodal emotion recognition, and adaptive question generation with transformer-based resume parsing for automated candidate evaluation.*	*Reference for adaptive question generation design; emotion detection is out of scope for the MVP but informs a possible future feature.*
https://cdn.openai.com/papers/whisper.pdf	*Introduces Whisper, trained on 680,000 hours of multilingual, multitask supervised data, showing strong robustness to accents, noise, and technical vocabulary compared to prior ASR systems.*	*Basis for the project's speech-totext component (via fasterwhisper), chosen for accuracy and free local deployment.*
https://arxiv.org/abs/2005.11401	*Introduces combining a retriever with a generator model, jointly fine-tuned, outperforming parametric-only and prior retrieve-and-read approaches.*	*Conceptual basis for grounding LLM-generated questions in the candidate's specific resume content.*
https://link.springer.com/article/10.1007/s00521-020-05351-2	*Trains a neural network on a small seed set of labeled resume education sections and improves predictions on unlabeled data via a correction module.*	*Highlights the difficulty of parsing unstructured resume sections, supporting the decision to rely on general-purpose LLMs rather than narrow custom trained parsers for the MVP.*